



Smart Automated Car Wash Management System: AutoWash Pro with Advanced Booking & Loyalty Program

Smart Automated Car Wash Management System with Advanced Booking & Loyalty Program

(AutoWash Pro)

Version Information

Item	Content
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1. Introduction

1.1 Purpose

The purpose of this document is to describe the functional and non-functional requirements for the Smart Automated Car Wash Management System with Advanced Booking and Loyalty Program (AutoWash Pro).

This system is designed to:

- automate car wash booking processes
- improve customer retention through a loyalty system
- support tier-based customer benefits
- provide promotional management for administrators
- generate operational reports

This document serves as a reference for:

- developers
 - testers
 - project managers
 - instructors
 - stakeholders
-

1.2 Scope

AutoWash Pro is a web and mobile-based management system for automated car wash services.

The system provides:

- customer registration and authentication
- vehicle and profile management
- advance booking with priority scheduling
- loyalty point accumulation
- membership tier management
- reward redemption
- promotion management
- reporting and analytics

The project includes:

- Web Application for Admin
 - Mobile/Web Application for Customers
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1.3 Definitions, Acronyms, and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
LPR	License Plate Recognition
CRUD	Create, Read, Update, Delete
SQL	Structured Query Language
UI	User Interface
API	Application Programming Interface
RTO	Recovery Time Objective

1.4 Intended Audience

This document is intended for:

- software developers
- testers

- database designers
 - project supervisors
 - project team members
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1.5 References

- IEEE Software Requirements Specification Standard
 - PRJ301 Project Description
 - Java Servlet/JSP Documentation
 - SQL Server Documentation
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2. Overall Description

2.1 Product Perspective

AutoWash Pro is a standalone smart car wash management system integrated with:

- booking management
- loyalty engine
- customer relationship management
- promotional management

The system uses:

- Java Servlet/JSP for backend and frontend
 - SQL Server for database management
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2.2 Product Functions

The system supports the following major functions:

Customer Functions

- Register account
- Login/logout
- Manage customer profile
- Manage vehicle information

- Create booking
- View booking history
- View loyalty points
- Redeem rewards

Admin Functions

- Manage customer accounts
 - Configure loyalty rules
 - Configure membership tiers
 - Manage promotions
 - View reports and analytics
-

2.3 User Classes and Characteristics

Customer

Characteristics:

- Car owners
- Uses mobile/web application
- Requires easy booking experience

Permissions:

- register/login
 - booking
 - reward redemption
 - view loyalty information
-

Admin

Characteristics:

- System administrator or manager
- Has full system access

Permissions:

- manage loyalty configuration

- manage promotions
 - view reports
 - manage customer data
-

2.4 Operating Environment

Component	Technology
Backend	Java Servlet (JDK 8)
Frontend	JSP, HTML, CSS
Database	SQL Server
Application Server	Apache Tomcat
OS	Windows
Application Server	Apache Tomcat 9
Development IDE	Apache NetBeans 13

2.5 Design and Implementation Constraints

- Must use Java Servlet/JSP
 - Must use SQL Server
 - Must support role-based authentication
 - Must support responsive UI
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2.6 Assumptions and Dependencies

- Users have internet connection
 - SQL Server is available
 - LPR camera integration may require third-party hardware
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3. External Interface Requirements

3.1 User Interfaces

The system shall provide:

- login screen
 - registration screen
 - booking dashboard
 - loyalty dashboard
 - admin management dashboard
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3.2 Hardware Interfaces

The system may integrate with:

- LPR camera devices
 - barcode/QR scanners
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3.3 Software Interfaces

Software	Purpose
SQL Server	Database
Apache Tomcat	Application deployment
JavaMail API	Email notification

3.4 Communication Interfaces

The system shall use:

- HTTP/HTTPS protocols
 - RESTful communication (optional future enhancement)
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4. System Features / Functional Requirements

4.1 Authentication Module

FR-01 Customer Registration

Description

The system shall allow customers to create new accounts.

Inputs

- Full name
- Phone number
- Password
- License plate

Processing

- Validate information
- Check duplicate phone number
- Save customer record

Outputs

- Successful registration message

Priority

High

FR-02 Login

Description

The system shall allow users to login using phone number and password.

Preconditions

User account exists.

Outputs

- Customer dashboard
 - Admin dashboard
-

FR-03 Logout

The system shall allow authenticated users to logout securely.

4.2 Customer Profile Module

FR-04 View Profile

Customers can view:

- personal information
 - vehicle information
 - loyalty tier
 - point balance
 - next reward
 - total spend
 - total visits
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FR-05 Update Profile

Customers can update:

- name
 - password
 - vehicle information
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4.3 Booking Module

FR-06 Create Booking

Description

Customers can book car wash services in advance.

Inputs

- service type
- booking date
- booking time

Business Rules

Booking window depends on customer tier.

Outputs

Booking confirmation.

FR-07 Booking History

Customers can view:

- booking history
 - payment history
 - completed services
-

FR-08 Priority Queue

Higher membership tiers receive earlier booking priority.

4.4 Loyalty Module

FR-09 Point Accumulation

Customers earn points after successful payment.

Rule

Rule: Point calculation must follow BR-03 (Tier-based bonus points)

FR-09A Spending Tracking

System shall record total customer spending.

FR-09B Visit Tracking

System shall record completed wash visits.

FR-10 Tier Management

The system shall automatically upgrade or downgrade customer tiers monthly.

FR-11 Reward Redemption

Customers can redeem points for:

- discounts
 - free wash
 - free wax
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FR-12 Point Expiry

Loyalty points shall expire after 12 months.

4.5 Promotion Module

FR-13 Promotion Management

Admins can:

- create promotions
 - update promotions
 - delete promotions
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FR-14a Targeted Promotions

Admins can send promotions to:

- Silver customers
- Gold customers
- Platinum customers

FR-14b AI Recommendations:

The system shall use AI to recommend personalized promotions to customers.

4.6 Reporting Module

FR-15 Customer Reports

Admins can view:

- total customers
- active customers

- loyalty statistics
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FR-16 Revenue Reports

Admins can view:

- daily revenue
 - monthly revenue
 - booking statistics
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5. Non-Functional Requirements

NFR-01 Performance

The system shall respond within ≤ 5 seconds for 95% of requests.

NFR-02 Booking Processing

Booking transactions shall complete within ≤ 5 seconds.

NFR-03 Concurrent Users

The system shall support at least 500 concurrent users.

Concurrent users = multiple users using the system simultaneously.

NFR-04 LPR Processing

License Plate Recognition shall identify vehicles within ≤ 10 seconds.

NFR-05 Availability

The system shall maintain 99.5% uptime excluding maintenance periods.

NFR-06 Backup and Recovery

The system shall:

- backup data daily
 - recover system within ≤ 2 hours
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NFR-07 Security

The system shall:

- encrypt passwords
 - prevent unauthorized access
 - support role-based access control
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NFR-08 Usability

The user interface shall be simple and responsive.

6. Business Rules

BR-01 Membership Tier Rules

Tier	Requirement
Member	Register + 1 wash
Silver	5 washes OR 2M VND
Gold	15 washes OR 6M VND
Platinum	30 washes OR 15M VND

BR-01 Membership Tier Rules

- **Member:** Requirement (Register + 1 wash) | Benefits (Base: 1 point = 1,000 VND spent)
- **Silver:** Requirement (5 washes OR 2M VND) | Benefits (+10% points, priority slot)
- **Gold:** Requirement (15 washes OR 6M VND) | Benefits (+20% points, free upgrade monthly)
- **Platinum:** Requirement (30 washes OR 15M VND) | Benefits (+30% points, free wash monthly)

BR-02 Booking Window Rules

Tier	Booking Window
Member	7 days
Silver	10 days
Gold	12 days
Platinum	14 days

BR-03 Loyalty Point Rules

Points are awarded based on membership tier. The base rate is 1 point for every 1,000 VND spent, with bonus percentages applied for Silver (+10%), Gold (+20%), and Platinum (+30%) members.

BR-04 Point Expiration Rule

Points expire after 12 months from earning date.

7. Use Case Specifications

UC-01 Register Account

Item	Description
Actor	Customer
Preconditions	User not logged in
Main Flow	Enter information → validate → save
Alternative Flow	Duplicate phone number
Postconditions	Account created

UC-02 Login

Item	Description
Actor	Customer/Admin
Preconditions	Account exists
Main Flow	Enter credentials → authenticate
Alternative Flow	Invalid credentials
Postconditions	Access dashboard

UC-03 Create Booking

Item	Description
Actor	Customer
Preconditions	Logged in
Main Flow	Select service → select slot → confirm
Alternative Flow	Slot unavailable
Postconditions	Booking created

UC-04 Redeem Reward

Item	Description
Actor	Customer
Preconditions	Enough points
Main Flow	Choose reward → confirm redemption

Alternative Flow	Insufficient points
Postconditions	Points deducted

UC-05 Manage Promotion

Item	Description
Actor	Admin
Preconditions	Admin logged in
Main Flow	Create/update/delete promotion
Postconditions	Promotion saved

8. Database Requirements

Main Entities

Entity	Description
Customer	Customer information
Vehicle	Vehicle information
Booking	Booking records
LoyaltyAccount	Loyalty data
Reward	Reward information
Promotion	Promotion information
Transaction	Payment records

Relationships

Relationship	Type
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Customer → Booking	One-to-Many
Customer → Vehicle	One-to-Many
Customer → LoyaltyAccount	One-to-One
Booking → Transaction	One-to-One

9. System Architecture Overview

Architecture Style

The system follows a 3-layer architecture:

Layer	Responsibility
Presentation Layer	JSP/HTML/CSS
Business Layer	Servlet/Service
Data Layer	SQL Server

10. Assumptions and Dependencies

- Internet connection is required
- SQL Server must be operational
- Users must register before booking
- LPR integration depends on external hardware

11. Appendices

Future Enhancements

- Mobile push notifications
- Online payment gateway
- Real-time booking tracking

- QR code check-in system
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End of Document